

MTBF Calculate Forms

Products Module: MR483000SG3-EF

Calulation Base on: MLT-HDBK-217F(Notice 2)

Applaied Environment;Good ground level

Environment temprature:25°C

Series NO	Name of Elements/Devices	Quantity	Qlty Coefficient	General failure ratio	II Q.λG	NλGIFQ
		N	πQ	λG(10E-6)	1.00E-05	-1.00E-05
1	Ceramic Capacitor	196	1	0.002	0.002	0.392
2	Metallic piastic Capacitor	19	1	0.002	0.002	0.038
3	Aluminum electrolytic Capacitor	14	1	0.00012	0.00012	0.00168
4	General Purpose Analog	28	1	0.001	0.0036	0.028
5	Fast recovery diode	12	1	0.001	0.001	0.012
6	Schoytty diode	13	1	0.025	0.025	0.325
7	zener diode	2	1	0.002	0.002	0.004
8	Transistor	21	1	0.000415	0.000415	0.008715
9	FUSE	2	1	0.01	0.01	0.02
10	Connector	1	1	0.0019	0.0019	0.0019
11	Inductor	5	1	0.00018	0.00018	0.0009
12	MOSFET	15	1	0.012	0.012	0.18
13	Retifier bridge	2	1	0.0028	0.0028	0.0056
14	Opto-cuupler	4	1	0.027	0.027	0.108
15	Relay	2	1	0.029	0.029	0.058
16	bidirectional triode thyristor	0	1	0.005	0.005	0.000
17	Metallized resistor	325	1	0.00086	0.00086	0.2795
18	Integrate circuit	12	1	0.0057	0.0057	0.0684
19	Voltage Regulator	4	1	0.00001	0.00001	0.00004
20	NTC resistor	2	1	0.000015	0.000015	0.00003
21	Operational Amplifiers	4	1	0.00033	0.00033	0.00132
22	Wire resistor	2	1	0.00016	0.00016	0.00032
23	Varistor	5	1	0.000015	0.000015	0.000075
24	transformer	3	1	0.0035	0.0035	0.0105
25	Solder dot	1905	1	0.0000069	0.0000069	0.0131445
26	fan	1	1	0.003	0.003	0.003
27	PCBA Pannel	3	1	0.0013	0.0013	0.0039
	Total	2602				1.5640245

$\lambda_s = \lambda_G \pi_Q \pi_S \pi_T \pi_E$	λG	πQ	πS	πT	πE
1.5640245	1.564	1	1	1	1

Actual MTBF	639376.17	Hours
Requirement MTBF	500000	Hours
Conclusion	PASS	